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Course: CSE411 – Artificial Intelligence

Question 1(a) [6 Marks] [CO3]

Question: A small weather station uses a logical agent to decide whether to turn on the air conditioner. Consider the following scenario as a KB. Now, Construct the following scenario into propositional logic, then use a full truth table to show whether the conclusion follows from the KB satisfiable or unsatisfiable:

- i) Today it is cloudy.
- ii) Today humidity is low.
- iii) If it is cloudy or humid, then the weather is uncomfortable.
- iv) If the weather is uncomfortable, then the air conditioner is on.

Solution 1(a):

Step 1: Define Propositional Symbols

Let:

- **c** = Today it is cloudy
- **h** = Today humidity is low (humid)
- **u** = The weather is uncomfortable
- **a** = The air conditioner is on

Step 2: Convert to Propositional Logic

Statement	Propositional Form
i) Today it is cloudy	c
ii) Today humidity is low	h
iii) If cloudy or humid → uncomfortable	$(c \vee h) \rightarrow u$

iv) If uncomfortable → AC is on u → a

KB = { c, h, (c ∨ h) → u, u → a }

Conclusion (α) = a (The air conditioner is on)

Step 3: Full Truth Table

We have 4 variables: c, h, u, a → 2^4 = 16 rows

c	h	u	a	c∨h	(c∨h)→u	u→a	KB satisfied?	α = a
F	F	F	F	F	T	T	❌ (c=F)	F
F	F	F	T	F	T	T	❌ (c=F)	T
F	F	T	F	F	T	F	❌ (c=F)	F
F	F	T	T	F	T	T	❌ (c=F)	T
F	T	F	F	T	F	T	❌ (h=T but (c∨h)→u fails)	F
F	T	F	T	T	F	T	❌	T
F	T	T	F	T	T	F	❌ (c=F)	F
F	T	T	T	T	T	T	❌ (c=F)	T
T	F	F	F	T	F	T	❌ ((c∨h)→u fails)	F
T	F	F	T	T	F	T	❌	T
T	F	T	F	T	T	F	❌ (u→a fails)	F
T	F	T	T	T	T	T	❌ (h=F)	T
T	T	F	F	T	F	T	❌ ((c∨h)→u fails)	F
T	T	F	T	T	F	T	❌	T
T	T	T	F	T	T	F	❌ (u→a fails)	F
T	**T**	**T**	**T**	**T**	**T**	**T**	✅	**T**

Step 4: Analysis

The KB is satisfied **only in row 16** where c=T, h=T, u=T, a=T.

In **every row where KB is satisfied, α (a) = T**.

✅ Conclusion: KB ⊨ α — The conclusion is SATISFIABLE. The air conditioner is ON, and the KB entails α.

Question 1(b) [4 Marks]

Question: Assume that KB = (p → q) ∧ r, r ↔ s, p ∨ s and α = (q ∧ ¬r) → s. Find whether KB ⊨ α.

Solution 1(b):

Step 1: Identify the components

KB consists of:

1. (p → q) ∧ r
2. r ↔ s
3. p ∨ s

α = (q ∧ ¬r) → s

Variables: p, q, r, s → 2^4 = 16 rows

Step 2: Truth Table

p	q	r	s	p→q	(p→q)∧r	r↔s	p∨s	KB holds?	q∧¬r	(q∧¬r)→s	α holds?
F	F	F	F	T	F	T	F	❌	F	T	T
F	F	F	T	T	F	F	T	❌	F	T	T
F	F	T	F	T	T	F	F	❌ (r↔s fails)	F	T	T
F	F	T	T	T	T	T	T	❌ (p∨s=T✅, but p→q: F→F=T✅) — check p∨s: s=T ✅ — **KB** ✅	F	T	**T**
F	T	F	F	T	F	T	F	❌	T	F	F

F	T	F	T	T	F	F	T	❌		T	T	T
F	T	T	F	T	T	F	F	❌		F	T	T
F	T	T	T	T	T	T	T	✅		F	T	**T**
T	F	F	F	F	F	T	T	❌		F	T	T
T	F	F	T	F	F	F	T	❌		F	T	T
T	F	T	F	F	F	F	T	❌		F	T	T
T	F	T	T	F	F	T	T	❌		F	T	T
T	T	F	F	T	F	T	T	❌		T	F	F
T	T	F	T	T	F	F	T	❌		T	T	T
T	T	T	F	T	T	F	T	❌ (r↔s fails)		F	T	T
T	T	T	T	T	T	T	T	✅		F	T	**T**

Step 3: Rows where KB is satisfied:

- **Row 4:** p=F, q=F, r=T, s=T → α = T ✅
- **Row 8:** p=F, q=T, r=T, s=T → α = T ✅
- **Row 16:** p=T, q=T, r=T, s=T → α = T ✅

✅ Conclusion: In ALL rows where KB is satisfied, α is also TRUE. Therefore, KB ⊨ α holds.

Question 2(a) [4 Marks] [CO3]

Question: Imagine two of your friends, Amin(alpha) and Samin(beta) are playing a tic-tac-toe game. You want Samin to win the game. Figure 1 is the tree of their game. Now, Develop solution on your known memory efficient gaming algorithm to find the result state.

Solution 2(a):

Algorithm: Alpha-Beta Pruning (Memory Efficient Gaming Algorithm)

Alpha-Beta Pruning is the memory-efficient version of Minimax. It prunes branches that cannot affect the final decision, saving computation and memory.

Since we want Samin (beta) to win → Samin is the MINIMIZER, Amin is the MAXIMIZER.

Wait — in standard game tree terms:

- **Amin (alpha) = MAX player** (tries to maximize score)
- **Samin (beta) = MIN player** (tries to minimize score)

We want **Samin to win**, so we apply **Alpha-Beta Pruning** from MIN's perspective.

Alpha-Beta Pruning Rules:

- **α (alpha):** Best value MAX can guarantee (starts at -∞)
- **β (beta):** Best value MIN can guarantee (starts at +∞)
- **Prune** when α ≥ β

Reading the Leaf Values from Figure 1 (left to right):

-19, -2, -4, -4, -12, 2, 0, -15, -4, -7, 17, -16, 14, 4, 11, 16, 17, 13, 4, 1, 14, -3, -4, 19, 9, -2, 10

Tree Structure Analysis:

The tree has:

- **Level 0 (Root):** MAX node (Amin)
- **Level 1:** MIN nodes (Samin)
- **Level 2:** MAX nodes (Amin)
- **Level 3:** Leaf nodes

Alpha-Beta Pruning Trace:

Left subtree of root (Node A - MIN):

Left child of A (MAX node, Node D):

- Leaves: -19, -2, -4 → MAX(-19, -2, -4) = **-2**
- α updated to -2

Middle-left child of A (MAX node, Node E):

- Leaves: -4, -12, 2 → MAX(-4, -12, 2) = **2**

Right of A (MAX node, Node F):

- Leaves: 0, -15, -4 → MAX(0,-15,-4) = **0**

MIN(A) = MIN(-2, 2, 0) = -2

Middle subtree of root (Node B - MIN):

Left child of B (MAX node, Node G):

- Leaves: -7, 17 → MAX(-7, 17) = **17**
- β = -2 (from root), now child gives 17 > β → **PRUNE remaining** if applicable

Actually tracing carefully:

- Node G: leaves -7, 17 → MAX = 17
- Node H: leaves -16, 14, 4 → MAX = 14
- Node I: leaves 11, 16, 17 → MAX = 17

MIN(B) = MIN(17, 14, 17) = 14

Right subtree of root (Node C - MIN):

Left child of C (MAX node, Node J):

- Leaves: 13, 4, 1 → MAX = **13**

Middle child of C (MAX node, Node K):

- Leaves: 14, -3, -4 → MAX = **14**

Right child of C (MAX node, Node L):

- Leaves: 19, 9, -2, 10 → MAX = **19**

MIN(C) = MIN(13, 14, 19) = 13

Root (MAX node):

MAX(A, B, C) = MAX(-2, 14, 13) = 14

✅ Result State: The optimal value is 14, achieved through Node B (middle branch).

Alpha-Beta Pruning successfully identified the best path without exploring all nodes, saving memory and time compared to pure Minimax.

Question 2(b) [4 Marks]

Question: Consider the Question 2a and now you want Amin to win the game. Will the game tree and final state be the same now? Illustrate your answer for the alternate gaming algorithm mentioned in question 2a. Compare the performance measures of the two algorithms.

Solution 2(b):

When Amin (MAX) needs to win:

Now we apply **pure Minimax Algorithm** (the alternate algorithm to Alpha-Beta Pruning).

- **Amin = MAX player** → wants to maximize
- **Samin = MIN player** → wants to minimize

Minimax Trace (same tree values):

Using same computed values:

- MIN(A) = -2
- MIN(B) = 14
- MIN(C) = 13

Root MAX = MAX(-2, 14, 13) = 14

Will the game tree and final state be the same?

Yes — the final result (14) is the same. Both Minimax and Alpha-Beta Pruning always produce the **same optimal result**. The difference is only in **how they compute it**.

The game tree structure remains the same. The final state value = **14** in both cases.

Performance Comparison Table:

Performance Measure	Minimax Algorithm	Alpha-Beta Pruning
Algorithm Type	Exhaustive Search	Pruned Search
Time Complexity	O(b^m)	O(b^(m/2)) best case
Space Complexity	O(bm)	O(bm)
Nodes Explored	ALL nodes explored	Many nodes pruned/skipped

Optimality	Always optimal	Always optimal (same result)
Memory Efficiency	Less efficient	More efficient (fewer nodes)
Pruning	No pruning	Prunes α ≥ β branches
Best Case	O(b^m)	O(b^(m/2))
Worst Case	O(b^m)	O(b^m)

Where **b** = branching factor, **m** = maximum depth of tree.

✅ Conclusion:

- Both algorithms give the **same optimal result = 14**
- Alpha-Beta Pruning is **more memory and time efficient**
- Minimax explores **every** node; Alpha-Beta **skips irrelevant** branches

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Question 2(c) [2 Marks]

Question: Explain the elements of search problem representing a game.

Solution 2(c):

A game as a search problem consists of the following elements:

1. Initial State (S_0): The starting configuration of the game board. Example: empty 3×3 tic-tac-toe board at the beginning.

2. Players: A function that defines which player (MAX or MIN) has the move in a given state. Example: Amin (MAX) and Samin (MIN).

3. Actions: The set of legal moves available to the current player in a given state. Example: placing X or O in any empty cell.

4. Transition Model (Result): A function Result(s, a) that defines the resulting state after a player takes action 'a' in state 's'.

5. Terminal Test: A function that checks whether the game is over (win, loss, or draw). It identifies terminal states.

6. Utility Function (Payoff Function): A numeric value assigned to terminal states indicating the outcome for a player. Example: +1 for win, -1 for loss, 0 for draw.

Question 3(a) [3 Marks] [CO2]

Question:

Scenario 1: A new grammar is proposed with the following rules:

- S → aSBC | aBC
- CB → BC
- aB → ab
- bB → bb
- bC → bc
- cC → cc

Scenario 2: Another Context Free Grammar is proposed:

- S → AB
- A → aA | ε
- B → bB | ε

Classify this grammar within the Chomsky hierarchy (Type-0 to Type-3) and justify your answer.

Solution 3(a):

Chomsky Hierarchy Recap:

Type	Name	Rule Constraint
Type-0	Unrestricted Grammar	No restriction
Type-1	Context-Sensitive Grammar (CSG)	
Type-2	Context-Free Grammar (CFG)	A → β, where A is a single non-terminal
Type-3	Regular Grammar	A → aB or A → a (right linear)

Classification of Scenario 1:

Examining the rules:

- **S → aSBC | aBC** — Left side is single non-terminal ✅ (CFG-like)
- **CB → BC** — Left side has TWO symbols (context-sensitive rule)

- **aB → ab** — Left side: terminal + non-terminal (two symbols)
- **bB → bb** — Left side: two symbols
- **bC → bc** — Left side: two symbols
- **cC → cc** — Left side: two symbols

Rules like **CB → BC, aB → ab** have more than one symbol on the left side, so this is **NOT CFG (Type-2)**.

Check if it is Type-1 (CSG): All rules satisfy |α| ≤ |β|:

- CB → BC: |2| ≤ |2| ✅
- aB → ab: |2| ≤ |2| ✅
- bB → bb: |2| ≤ |2| ✅
- bC → bc: |2| ≤ |2| ✅
- cC → cc: |2| ≤ |2| ✅

✅ **Scenario 1 is Type-1: Context-Sensitive Grammar (CSG)**

Classification of Scenario 2:

- **S → AB** — single non-terminal on left ✅
- **A → aA | ε** — single non-terminal on left ✅
- **B → bB | ε** — single non-terminal on left ✅

Every rule has exactly ONE non-terminal on the left side → **CFG (Type-2)**

Can it be Type-3 (Regular)?

- **A → aA** is right-linear ✅
- **B → bB** is right-linear ✅
- **S → AB** — has TWO non-terminals on the right ❌ (not regular)

✅ **Scenario 2 is Type-2: Context-Free Grammar (CFG)**

Question 3(b) [2 Marks]

Question: From Scenario 2: Convert the grammar to CNF, now Discuss that the derived language remains unchanged.

Scenario 2 Grammar:

- S → AB
- A → aA | ε
- B → bB | ε

Solution 3(b):

Chomsky Normal Form (CNF) Rules:

Every production must be either:

1. **A** → **BC** (two non-terminals)
2. **A** → **a** (single terminal)

Step 1: Handle ε-productions

ε-producing non-terminals: A → ε and B → ε

Remove ε-productions and add alternatives:

- A → aA | ε becomes **A** → **aA | a** (when A is removed, aA becomes a)
- B → bB | ε becomes **B** → **bB | b**
- S → AB: since A and B can be ε, add: **S** → **AB | A | B | ε**

But to keep language the same, we handle nullable variables:

Updated grammar:

- S → AB | A | B
- A → aA | a
- B → bB | b

Step 2: Convert to CNF

S → **AB** ✅ (already CNF form) **S** → **A** — unit production, substitute: **S** → **aA | a S** → **B** — unit production, substitute: **S** → **bB | b**

A → **aA:** introduce **T_a** → **a**, then **A** → **T_a A** ✅ **A** → **a** ✅ already CNF

B → **bB:** introduce **T_b** → **b**, then **B** → **T_b B** ✅ **B** → **b** ✅ already CNF

Final CNF Grammar:

- S → AB | T_a A | a | T_b B | b
- A → T_a A | a
- B → T_b B | b
- T_a → a
- T_b → b

Discussion: Language Remains Unchanged

The original grammar generates the language: **L = { $a^n b^m$ | n ≥ 0, m ≥ 0 }** (strings of zero or more a's followed by zero or more b's, including ε)

The CNF version generates the **same language** because:

1. ε-removal was compensated by adding nullable alternatives to S
2. Unit productions were eliminated by direct substitution (no new strings added/removed)
3. Terminal-with-non-terminal rules were split using new non-terminals (T_a, T_b) — these only rename, not change derivation

✅ **Therefore, the CNF transformation preserves the original language $L = \{a^n b^m \mid n,m \geq 0\}$.**

Question 4(a) [3 Marks] [CO2]

Question: In a manufacturing company, four distinct machines denoted as M_1, M_2, M_3, and M_4 are responsible for producing widgets. M_1 produces 10% of the total widgets, M_2 produces 20%, M_3 produces 30%, and M_4 produces 40%. The defective rates for the respective machines are 4%, 6%, 2% and 5%. If a randomly selected widget is found not to be defective, Identify the probability that it was produced by M_3.

Solution 4(a):

Given Information:

Machine	$P(M_i)$	$P(D\|M_i)$	$P(D'\|M_i) = 1 - P(D\|M_i)$
M_1	0.10	0.04	0.96
M_2	0.20	0.06	0.94
M_3	0.30	0.02	0.98
M_4	0.40	0.05	0.95

Find: $P(M_3 \mid D')$ — probability widget came from M_3 given it is NOT defective

Applying Bayes' Theorem:

$$P(M_3 | D') = \frac{P(D' | M_3) \cdot P(M_3)}{P(D')}$$

Step 1: Calculate P(D') using Total Probability Theorem:

$P(D') = P(D'|M_1)\cdot P(M_1) + P(D'|M_2)\cdot P(M_2) + P(D'|M_3)\cdot P(M_3) + P(D'|M_4)\cdot P(M_4)$

$P(D') = (0.96 \times 0.10) + (0.94 \times 0.20) + (0.98 \times 0.30) + (0.95 \times 0.40)$

$P(D') = 0.096 + 0.188 + 0.294 + 0.380$

$P(D') = 0.958$

Step 2: Calculate Numerator:

$P(D'|M_3) \cdot P(M_3) = 0.98 \times 0.30 =$ **0.294**

Step 3: Apply Bayes' Theorem:

$$P(M_3 | D') = \frac{0.294}{0.958} = 0.3069$$

✅ Answer: $P(M_3 | D') \approx 0.3069$ or 30.69%

There is approximately a **30.69% probability** that a non-defective widget was produced by M_3.

Question 4(b) [2 Marks]

Question: Explain the different reasons for uncertainty encountered in Artificial Intelligence systems and discuss how these uncertainties affect AI decision-making processes.

Solution 4(b):

Sources of Uncertainty in AI Systems:

1. Incomplete Information: AI agents often lack complete knowledge about the world. For example, a robot may not know what is behind a wall. This forces the agent to make decisions based on partial data, potentially leading to suboptimal choices.

2. Noisy/Incorrect Observations: Sensors and data sources may provide inaccurate or noisy readings. Example: a temperature sensor may give slightly wrong values, causing an AI to make wrong inferences.

3. Stochastic (Random) Environments: The world is non-deterministic. Even with the same action, outcomes can vary randomly (e.g., in weather prediction or stock markets), making it hard to predict future states.

4. Ignorance / Lack of Knowledge: AI may simply not have been trained on certain scenarios, leading to uncertain responses in novel situations.

5. Model Uncertainty: The AI model itself may be an imperfect approximation of reality. Assumptions made during modeling may not hold in real scenarios.

Effect on AI Decision-Making:

- AI must use **probabilistic reasoning** (e.g., Bayes' theorem) instead of definite conclusions

- Decisions become **risk-based** rather than certainty-based
- May require **multiple hypotheses** to be maintained simultaneously
- Can lead to **incorrect or suboptimal actions** if uncertainty is not properly handled
- Requires **confidence thresholds** for action (like the perceptron threshold θ in Q5)

Question 5(a) [5 Marks] [CO4]

Question Scenario: Daffodil International University (DIU) has implemented a Smart Authentication System to control access and track attendance across multiple zones such as classrooms, labs, and the library. The system uses biometric scans, RFID cards, and face recognition technology. The system uses a perceptron-based decision model to compute a confidence score based on three key inputs:

- x_1: Biometric match score (0 to 1)
- x_2: RFID card validity (1, 0)
- x_3: Face recognition score (0 to 1)
- Weights: (W1, W2, W3) = (0.5, 0.3, 0.2)
- Threshold (θ) = 0.7
- Grant access if weighted sum ≥ 0.7, else deny.

Question 5(a): A student attempts to enter the Laboratory with the following inputs:

- Biometric match = 0.8
- RFID validity = 1
- Face recognition score = 0.6

Now, Examine the weighted sum and determine if the student is granted access to the laboratory. Show all steps clearly.

Solution 5(a):

Given:

- x_1 = 0.8 (Biometric match)
- x_2 = 1 (RFID validity)
- x_3 = 0.6 (Face recognition)
- W_1 = 0.5, W_2 = 0.3, W_3 = 0.2
- Threshold θ = 0.7

Step 1: Perceptron Weighted Sum Formula:

$$\text{Weighted Sum} = W_1 \cdot x_1 + W_2 \cdot x_2 + W_3 \cdot x_3$$

Step 2: Substitute Values:

$$\text{Weighted Sum} = (0.5 \times 0.8) + (0.3 \times 1) + (0.2 \times 0.6)$$

$$= 0.40 + 0.30 + 0.12$$

$$= \mathbf{0.82}$$

Step 3: Apply Decision Rule:

Condition	Result
Weighted Sum ≥ θ	Grant Access ✅
Weighted Sum < θ	Deny Access ❌

0.82 ≥ 0.7 → Condition is TRUE

Step 4: Decision:

$$\text{Output} = \begin{cases} 1 \text{ (Grant Access)} & \text{if } \sum W_i x_i \geq \theta \\ 0 \text{ (Deny Access)} & \text{if } \sum W_i x_i < \theta \end{cases}$$

Since **0.82 ≥ 0.7**, Output = **1**

✅ Conclusion: The student IS GRANTED ACCESS to the laboratory.

The weighted confidence score of **0.82** exceeds the threshold of **0.7**, so the system logs attendance and grants access.

Question 5(b) [5 Marks] [CO4]

Question: Analyze how the system demonstrates the concepts of perception and computation in the decision-making process for accessing different zones at DIU. Also, mention one advantage and one challenge of using this automated system in the library and classrooms at DIU.

Solution 5(b):

Perception in the DIU Smart Authentication System:

Perception refers to the system's ability to gather information from the environment through sensors:

- **Biometric Scanner** → Perceives fingerprint/iris → produces x_1 (match score 0 to 1)
- **RFID Card Reader** → Perceives card signal → produces x_2 (valid=1, invalid=0)
- **Face Recognition Camera** → Perceives facial features → produces x_3 (recognition score 0 to 1)

These three inputs represent the **percept sequence** of the AI agent — raw data collected from the real-world environment. Just like a logical agent perceives its environment before acting, the system perceives identity features before making an access decision.

Computation in the DIU Smart Authentication System:

Computation refers to processing the perceived inputs to reach a decision:

The **Perceptron model** computes:

$$\text{Confidence Score} = W_1 x_1 + W_2 x_2 + W_3 x_3$$

This is the **inference/reasoning** step — the agent uses weights (knowledge about importance of each input) and applies a threshold function (activation function):

$$\text{Output} = \begin{cases} 1 \text{ (Access Granted)} & \text{if score} \geq 0.7 \ 0 \text{ (Access Denied)} & \text{if score} < 0.7 \end{cases}$$

This mimics how a biological neuron fires only when stimulation exceeds a threshold — hence "perceptron-based."

Different Zones — Computation Adapts:

- **Library:** May use lower threshold (more lenient access)
- **Lab:** Uses θ = 0.7 (moderate security)
- **Classrooms:** May have different weights depending on zone importance

One Advantage:

Automated Accuracy & Speed: The system eliminates manual attendance and reduces human error. It can simultaneously authenticate and log attendance in milliseconds across multiple zones (classrooms, labs, library) without any staff intervention, increasing efficiency and reliability.

One Challenge:

Biometric Data Privacy & System Failure Risk: Storing and processing sensitive biometric data (fingerprints, facial features) raises serious **privacy and security concerns**. If the system is hacked or fails due to power/network issues, access to critical zones like laboratories could be completely disrupted, affecting academic activities. Also, students with injuries or disabilities may face false rejections.

— End of All Solutions —